

Exercise 13

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Problem 1a. Let M be a Poisson Point Process on $(E, \mathcal{E})$ with a σ -finite intensity measure μ .

Solution. 1. True

2. False

3. False

4. False - μ can tell us about X_i .

5. True

■

Problem 1b.

Solution. 1. False

2. False

3. False

4. True

5. False

■

Problem 2. Let M be a Poisson Point Process on $\mathbb{R}$ with $\mu = \text{Leb}_{\mathbb{R}}$. Consider the map $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$T(x) := \begin{cases} \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Claim 2a. Show that $M' = T\#M$ is a Poisson Point Process

Proof. We examine $T\#\mu$. Let $A = [1, \infty) \cup (-\infty, 1]$. $(T\#\mu)(A) = \mu(T^{-1}(A)) = \mu([-1, 1]) = 2$. Let $B_i = [\frac{1}{i+1}, \frac{1}{i}] \cup [-\frac{1}{i}, -\frac{1}{i+1}]$. $(T\#\mu)(B_i) = \mu(T^{-1}(B_i)) = \mu([i, i+1] \cup [-i-1, -i]) = 2$. $(\bigcup_{i \in \mathbb{N}} B_i) \cup A = \mathbb{R}$. Thus $T\#\mu$ is σ -finite. Thus $T\#M$ is a Poisson Point Process, by the Mapping Theorem. $\square$

Claim 2b. Determine its intensity.

Proof. 1. $\forall a \leq 0 < b$,

$$(T\#\mu)([a, b]) = \infty$$

2. $\forall 0 < a < b$

$$(T\#\mu)([a, b]) = \mu\left(\left[\frac{1}{b}, \frac{1}{a}\right]\right) = \frac{1}{a} - \frac{1}{b}$$

3. $\forall b < a < 0$

$$(T\#\mu)([b, a]) = \mu\left(\left[\frac{1}{a}, \frac{1}{b}\right]\right) = \frac{1}{b} - \frac{1}{a}$$

□

Problem 3. Let $M = \sum_{i=1} \delta_{X_i}$ be a Poisson point process on $\mathbb{R}^d$ with intensity $\mu = \text{Leb}(\mathbb{R}^d)$. Let us consider $(R_i)_i$ a sequence of i.i.d random variables independent from M with law ρ . Let $O = \cup_i B(X_i, R_i)$ where $B(X, R) \subset \mathbb{R}^d$ is the open ball centered at X of radius of R .

Claim 3a. Let M_0 be the number of balls $B(X_i, R_i)$ which contain the origin. Show M_0 is a well defined random variable.

Proof. Let

$$A = \{(x, r) \in \mathbb{R}^d \times [0, \infty) \mid |x| < r\}$$

Let $\overline{M} = \sum_{i=1} \delta_{(X_i, R_i)}$ be the R -marked process. By the marking theorem, $\overline{M}$ is a poisson point process with intensity $\mu \otimes \rho$.

$\overline{M}(A) = \#\{i \mid |X_i| < R_i\} = M_0$. Thus $M_0 = \overline{M}(A) \sim \text{Poisson}((\mu \otimes \rho)(A))$ by definition of poisson point process. Thus M_0 is a well defined random variable. □

Claim 3b. Show that the event $\{O = \mathbb{R}^d\}$ is measurable and $\mathbb{P}[O = \mathbb{R}^d] = 1$ if and only if $\int_0^\infty r^d \rho(dr) = \infty$

Proof. $\mathbb{R}^d$ is a separable space with a countable dense subset $\mathbb{Q}^d$. O is a union of open balls, and thus is a open set. So of $\mathbb{Q}^d \subseteq O \implies O = \mathbb{R}^d$ by definition of dense subset. Thus

$$\{O = \mathbb{R}^d\} = \{\mathbb{Q}^d \subseteq O\} = \bigcap_{q \in \mathbb{Q}^d} \{q \in O\}$$

The event $\{q \in O\}$ is saying that is $q \in B(X_i, R_i)$ for some i . Thus $\{q \in O\} = \bigcup_i \{|q - X_i| < R_i\}$ which is measurable.

Thus

$$\{O = \mathbb{R}^d\} = \{O = \mathbb{R}^d\} = \bigcap_{q \in \mathbb{Q}^d} \bigcup_i \{|q - X_i| < R_i\}$$

is measurable being the countable intersection of measurable events.

For $q \in \mathbb{R}^d$, $M_q := \#\{\text{balls covering } q\}$. By 3a, $M_q = \text{Poisson}((\mu \otimes \rho)(A_q))$ where $A_q = \{(x, r) \in \mathbb{R}^d \times [0, \infty) \mid |x - q| < r\}$. $(\mu \otimes \rho)(A_q) = \omega_d \int_0^\infty r^d \rho(dr)$ where ω_d is the volume of unit ball in $\mathbb{R}^d$.

$\implies$: Suppose $\Pr[O = \mathbb{R}^d] = 1$. That means $\forall x \in \mathbb{R}^d$, $M_x \geq 1$ a.s. Thus,

$$0 = \mathbb{P}[M_x = 0] = e^{-\omega_d \int_0^\infty r^d \rho(dr)} \implies \int_0^\infty r^d \rho(dr) = \infty$$

this is because ω_d is a constant.

$\Leftarrow$: Suppose $\int_0^\infty r^d \rho(dr) = \infty$. $\forall x \in \mathbb{R}^d$

$$\mathbb{P}[M_x = 0] = e^{-\omega_d \int_0^\infty r^d \rho(dr)} = 0$$

Thus $M_x \geq 1$ a.s. Thus $x \in O$ a.s. Thus $\mathbb{R}^d \subseteq O \implies O = \mathbb{R}^d$ a.s. $\square$

Problem 4. Let M be a poisson point process on $(E, \mathcal{E})$ with intensity measure μ . Recall that $\mathcal{L}_M$, Laplace Functional of N is given by:

$$\mathcal{L}_M(u) = E \left[\exp \left(- \int_E u(x) M(dx) \right) \right]$$

for all $u : E \rightarrow \mathbb{R}^+$ measurable.

Claim 4a. Let $B \in \mathcal{E}$ show that if $\mu(B) < \infty$ then

$$\mu(B) = -\frac{d}{dt} \mathcal{L}_M(t1_B)|_{t=0}$$

Proof. Suppose $B \in \mathcal{E}$ such that $\lambda = \mu(B) < \infty$.

$$\begin{aligned} \frac{d}{dt} (\mathcal{L}_M(t1_B)) &= \frac{d}{dt} \left(E \left[\exp \left(- \int_E t1_B M(dx) \right) \right] \right) \\ &= \frac{d}{dt} \left(E \left[\exp \left(-t \int_B M(dx) \right) \right] \right) \\ &= \frac{d}{dt} (E [\exp(-tM(B))]) \\ &= \frac{d}{dt} \exp(\lambda(e^{-t} - 1)) \end{aligned}$$

The last equality comes from the definition of a poisson point process.

$$-\frac{d}{dt} (\mathcal{L}_M(t1_B)) = -(-\lambda \exp(\lambda(e^{-t} - 1))e^{-t})$$

Thus,

$$-\frac{d}{dt} (\mathcal{L}_M(t1_B))|_{t=0} = \lambda = \mu(B)$$

$\square$

Claim 4b. Let $B \in \mathcal{E}$, note $\mu(B)$ doesn't need to be finite, then

$$\mathbb{P}[M(B) = 0] = \lim_{t \rightarrow \infty} \mathcal{L}_M(t1_B)$$

Proof. Note

$$L_M(t1_B) = E[\exp(-tM(B))]$$

as proven above.

1. $\mu(B) < \infty$: Then $L_M(t1_B) = \exp(\mu(B)(e^{-t} - 1))$. Thus

$$\lim_{t \rightarrow \infty} L_M(t1_B) = e^{-\mu(B)} = \frac{\mu(B)^0}{0!} e^{-\mu(B)} = \mathbb{P}[M(B) = 0]$$

2. $\mu(B)$ is not finite. Then $M(B) = \infty$ a.s. Thus $L_M(t1_B) = 0$ a.s for all $t > 0$. Thus

$$\lim_{t \rightarrow \infty} L_M(t1_B) = 0 = \mathbb{P}[M(B) = 0]$$

□

Problem 5. Let N be a poisson point process on $(E, \mathcal{E})$ with intensity measure μ , and let $u : E \rightarrow \mathbb{R}$ be measurable. Show that $\int u(x)N(dx)$ is a well defined random variable and that if $u \geq 0$ or $\int |u(x)|\mu(dx) < \infty$, then

$$E \left[\int u(x)N(dx) \right] = \int u(x)\mu(dx)$$

Claim 5a. $\int u(x)N(dx)$ is a random variable.

Proof. For a simple function $u = \sum_{i=0}^n c_i 1_{A_i}$ with disjoint measurable sets A_i .

$$\int u(x)N(dx) = \sum_{i=0}^n c_i N(A_i)$$

each $N(A_i)$ is a random variable, thus their linear combination is a random variable.

For any nonnegative function u , approximate it with a sequence of monotonic simple functions $(u_n)_{n \in \mathbb{N}}$. By MCT $\int u_n(x)N(dx) \rightarrow \int u(x)N(dx)$, almost surely. Since $\forall n \in \mathbb{N}$, $\int u_n(x)N(dx)$ is a random variable, $\int u(x)N(dx)$ is a random variable.

Let $u = u^+ - u^-$ where $u^+, u^- \geq 0$. By the above, $\int u^+(x)N(dx)$ and $\int u^-(x)N(dx)$ are random variables and thus the difference u is a random variable. □

Claim 5b. If $u \geq 0$ or $\int |u(x)|\mu(dx) < \infty$, then

$$E \left[\int u(x)N(dx) \right] = \int u(x)\mu(dx)$$

Proof. Let $u = \sum_{i=0}^n c_i 1_{A_i}$ for disjoint measurable sets A_i .

$$\begin{aligned} E \left[\int u(x) N(dx) \right] &= E \left[\sum_{i=0}^n c_i N(A_i) \right] \\ &= \sum_{i=0}^n c_i \mu(A_i) \\ &= \int u(x) \mu(dx) \end{aligned}$$

Let $u \geq 0$. Approximate u by a sequence of nondecreasing simple functions $(u_n)_{n \in \mathbb{N}}$. Thus by monotonic convergence theorem

$$\begin{aligned} \int u(x) N(dx) &= \lim_{n \rightarrow \infty} \int u_n(x) N(dx) \\ &= \lim_{n \rightarrow \infty} \int u_n(x) \mu(dx) \\ &= \int u(x) \mu(dx) \end{aligned}$$

Let u be a function such that $\int |u(x)| \mu(dx) < \infty$. Thus let $u^+, u^- \geq 0$ where $u = u^+ - u^-$. Thus $\int u^+(x) \mu(dx), \int u^-(x) \mu(dx) < \infty$.

$$\begin{aligned} E \left[\int u(x) N(dx) \right] &= E \left[\int u^+(x) N(dx) - \int u^-(x) N(dx) \right] \\ &= \int u^+(x) \mu(dx) - \int u^-(x) \mu(dx) \\ &= \int u(x) \mu(dx) \end{aligned}$$

□